



TEAM PROPOSAL

IDEALIZE 2026 - Team Proposal



Team Name	Algostrom
Category (web/app)	WebApp
School / Open	Open
Name of the web / app	SoundScout
Field of the web / app	AV Rentals

***Before completing the template please refer proposal guidelines document to understand the submission guidelines.**

Click here for the submission guidelines :

[Proposal Guideline](#)



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Problem & Solution

The Problem Statement

State the problem your application aims to solve and explain why it is important to address.

Application Blueprint

Provide an overview of your web/mobile application and describe its main features and functionalities.

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Innovation & Technical Design

Agentic AI Intergration

Explain how AI is incorporated into your solution and how it contributes to solving the identified problem.

Technical Architecture & Feasibility

Describe the technologies, tools, and system design of your application. Include a brief development plan demonstrating the feasibility of implementation.

***In the School Category, participants will be evaluated on Conceptual Design & App Logic instead of Agentic AI Integration.**

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Market Potential & Sustainability

Market Potential & Sustainability

Explain your target users, the value your solution offers, and its potential for long-term growth and impact.

Sustainability & Long- Term Impact

Describe how your solution can remain relevant, scalable, and impactful over time.

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Team Details

Please provide the necessary details of your team. All fields including photographs are required.



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Team Details

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The Problem Statement

Event organizers frequently struggle to determine the technical infrastructure required for their events. Whether it is a wedding, musical show, religious gathering, corporate conference, or community festival, organizers often lack the expertise to accurately estimate audio systems, lighting setups, stage requirements, visual displays, and power solutions.

In addition, the event rental industry lacks standardized pricing. There is no clear benchmark for equipment rental costs, making it difficult for organizers to determine whether a quotation is fair. As a result, some rental service providers charge significantly higher prices than the actual market value, while customers have limited information to compare options and negotiate effectively.

Consequently, organizers either depend on expensive technical consultants or make decisions based on guesswork, leading to inflated rental costs, equipment shortages, poor audience experiences, or even complete event failures. Meanwhile, rental providers face difficulties reaching potential customers and often spend considerable time preparing quotations for poorly defined requirements.

The event rental industry remains highly fragmented, with no centralized intelligent platform capable of translating event requirements into technically accurate infrastructure plans, providing pricing transparency, and connecting organizers with suitable service providers.

Application Blueprint

Track 1 – SoundScout AI Web Platform

A comprehensive web application where event organizers can describe their event using natural language or structured inputs such as event type, venue size, crowd count, location, and budget. SoundScout AI automatically generates a complete event infrastructure plan including audio systems, lighting systems, LED displays, stage requirements, and power solutions. The generated plan is then published to registered vendors who compete through a transparent bidding system.

Track 2 – Instant Rental & Emergency Response Platform

For urgent situations, users can instantly search and rent available equipment without waiting for bids. The system identifies nearby suppliers, checks real time equipment availability, and enables immediate booking. This mode is designed for emergencies such as equipment failures, last minute events, and unexpected capacity expansions.

Track 3 – Future AI Venue Intelligence

Users will be able to upload venue photographs, drone images, or floor plans. The AI will analyze spatial characteristics and generate recommended speaker placements, lighting arrangements, stage positioning, and equipment distribution plans for optimal event performance.

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Agentic AI Integration

SoundScout AI functions as a Virtual Event Infrastructure Consultant rather than a simple recommendation engine.

Powered by multimodal large language models and computer vision technologies, the AI agent can understand event descriptions, interpret venue images, reason about spatial constraints, estimate audience coverage requirements, and autonomously generate complete infrastructure plans.

The AI agent performs several coordinated tasks:

- Understands user requirements through natural language conversations.
- Estimates audio, lighting, visual display, staging, and power requirements.
- Generates structured equipment lists and technical specifications.
- Predicts fair market pricing ranges using historical quotation data.
- Matches event requests with suitable vendors based on inventory, location, availability, and reputation.
- Provides intelligent recommendations to help organizers make cost-effective decisions.

This transforms SoundScout AI from a rental marketplace into an autonomous digital event consultant accessible to anyone.

Technical Architecture

Frontend

- React.js
- Tailwind CSS
- Progressive Web App (PWA)

Backend

- Node.js
- Express.js

Database

- PostgreSQL

AI Layer

- Python
- Gemini API
- Scikit learn

Real-Time Services

- Firebase Cloud Messaging
- WebSockets

Computer Vision (Future)

- OpenCV
- YOLO
- Gemini Vision

Deployment

- Docker
- AWS

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Market Potential

- SoundScout AI targets a rapidly growing event services market consisting of multiple customer segments.
- Event Organizers – Weddings, concerts, festivals, school events, university events, religious programs, and corporate functions.
- Event Management Companies – Seeking faster technical planning and supplier sourcing.
- Rental Service Providers – Audio, lighting, LED wall, staging, and generator rental businesses seeking qualified leads.
- Educational Institutions – Schools and universities conducting large-scale events.
- Religious and Community Organizations – Frequently organizing public gatherings with limited technical expertise.
- By expanding beyond sound systems to complete event infrastructure planning, SoundScout AI addresses a significantly larger market opportunity and creates value for both organizers and suppliers.

Long Term Impact

- SoundScout AI aims to democratize professional event planning by making expert level infrastructure consultation accessible to everyone. The platform will increase transparency within the rental industry, reduce unnecessary event costs, improve event quality, create new opportunities for small and medium rental providers, and accelerate the digital transformation of the Sri Lankan events ecosystem. In the long term, SoundScout AI can evolve into a regional platform supporting complete event planning, infrastructure optimization, and intelligent venue design across South Asia.

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Sustainability

SoundScout AI establishes a sustainable two sided marketplace where event organizers gain access to intelligent infrastructure planning, while rental providers receive highly qualified, technically validated customer leads. This creates recurring value for both stakeholders and encourages long term platform engagement.

The platform leverages AI to reduce planning inefficiencies, minimize equipment underutilization, and optimize resource allocation across the event industry. By transforming traditionally offline rental processes into a data driven digital ecosystem, SoundScout AI creates lasting economic and operational benefits.

Financial sustainability is achieved through a hybrid business model consisting of transaction commissions, premium vendor memberships, sponsored listings, AI powered consultation services, and enterprise partnerships with event management companies. As the platform grows, accumulated event and equipment data will further enhance AI accuracy, creating a competitive advantage that strengthens over time.

By empowering organizers with expert level planning capabilities and enabling fair market access for rental providers, SoundScout AI fosters an inclusive, scalable, and future ready event services ecosystem capable of supporting thousands of events annually.



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THANK YOU